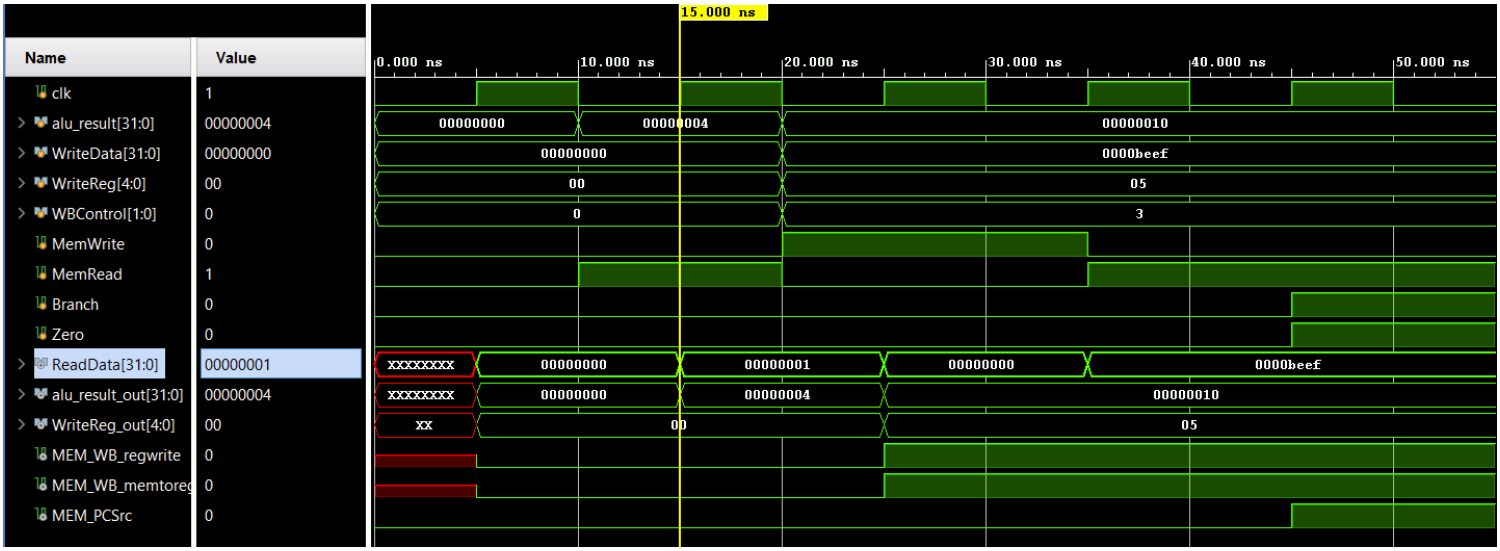
Brendan Han
Cheyenne Valencia

Memory Stage Testbench Waveform



Testing Memory Read
@ 10 ns - alu_result value = 00000004
MemRead signal is high
data.txt file is read and displays the dmem[1] value 00000001 at object ReadData[31:0]

Testing Memory Write
@ 20 ns - alu_result value = 00000010
MemWrite signal is high
Writing the value 0000BEEF to memory
Value 0000BEEF is now at address 00000010

Testing Memory Read after Write
@ 35 ns - alu_result value = 00000010
MemRead signal is high
MemWrite signal is low
ReadData[31:0] outputs 0000BEEF

WriteReg is initially set to 0
Changing WriteReg = 5 with WBControl = 3
WBControl = 3: Register Write On, Write to Data Memory
Value 0000BEEF is now saved to Register 5